INCH-POUND
MIL-M-38510/108(USAF)
AMENDMENT 3
28 February 2000
SUPERSEDING
AMENDMENT 2
29 September 1986

MILITARY SPECIFICATION

Inactive for new design as of 10 July 1995

MICROCIRCUITS, LINEAR TRANSISTOR
ARRAYS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/108(USAF), dated 9 April 1976, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 2: Delete “Three” and substitute “Two”.

1.2, delete and substitute:
“1.2 Part number. The part number shall be in accordance with MIL-M-38510.”

PAGE 2

* 3.1, at the end of the paragraph add the following sentence:
“This slash sheet has been modified to allow the manufacturer to use the alternate die/fabrication requirements of paragraph A.3.2.2 of MIL-PRF-38535 or other alternative approved by the Qualifying Activity.

PAGE 4

TABLE I, delete the temperature coefficient of base emitter voltage test and substitute with the revised test below:
“

| Test | Symbol | Conditions | | Device type | Limits 1/ Min | Limits 1/ Max | Units |
|---|---|---|---|---|---|---|---|
| Temperature coefficient of base emitter voltage 4/ 8/ | $\frac{\Delta V_{BE}}{\Delta T}$ | $V_{CE}$ = 3 V | -55°C ≤ $T_A$ ≤ +25°C | 01 | -2.2 | -1.5 | mV/°C |
| | | | | 02 | -2.2 | -1.3 | |
| | | $I_E$ = -1 mA | -55°C ≤ $T_A$ ≤ +25°C | 01 | -2.2 | -1.5 | mV/°C |
| | | | | 02 | -2.2 | -1.3 | |

“

TABLE I, static forward current transfer ratio (beta), $h_{FE}$ at $I_C$ = 1 mA, -55°C ≤ $T_A$ ≤ +25°C; maximum limit column (device 02 only): Delete “300”.

PAGE 5

3.6, delete and substitute:
“3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.”


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TABLE II, delete and substitute:

"TABLE II. Electrical test requirements

| MIL-STD-883 test requirement | Class S Devices | Class B devices |
|---|---|---|
| Interim electrical parameters (method 5004) | 1 | 1 |
| Final electrical test parameters (method 5005) | 1*, (2, 3, 4)** | 1*, (2, 3, 4)** |
| Group A test requirements (method 5005) | 1, 2, 3, 4, 5, 6, 7, 9, 10, 11 | 1, 2, 3, 4 |
| Group C end point and group B Class S electrical parameters (method 5005) | 1, 2, 3 and Table IV delta limits | Table IV delta limits and limits |
| Additional electrical subgroups for group C periodic inspection | None | 5, 6, 7, 9, 10, 11 |

* PDA applies to subgroup 1 (see 4.3e)

** $\frac{\Delta V_{BE}}{\Delta T}$ and $\frac{\Delta \left| V_{BEQA} - V_{BEQB} \right|}{\Delta T}$ tests as specified in table III herein for group A, subgroups 2 and 3, and $f_t$ tests as specified in table III herein for group A, subgroup 4, are not required for final electrical tests (for device 02 only) but shall be performed for group A sample testing."

PAGE 6

*3.7, delete second sentence in its entirety. Add the following paragraph 3.7.1:

"3.7.1 Certification/compliance mark. The certification mark for device classes Q and V shall be a "QML" or "Q" as required in MIL-PRF-38535. For class Q product built in accordance with A.3.2.2 of MIL-PRF-38535 or other alternative approved by the Qualifying Activity, the "QD" certification mark shall be used in place of the "QML" or "Q" certification mark."

PAGE 24

TABLE III, tests 68 through 72, maximum limits column: Delete "300".

PAGE 26

TABLE III, tests 119 through 123, maximum limits column: Delete "-1.5" and substitute "-1.3". Tests 128 through 132, maximum limits column: Delete "300".

PAGE 27

TABLE III, tests 156 through 160, maximum limits column: Delete "-1.5" and substitute "-1.3".

PAGE 30

4.5.2, line 2 following cooled to: Add "within 10°C of their power stable condition,".



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TABLE IV, delete and substitute:

"TABLE IV. Groups B and C end point electrical parameters ($T_A$ = 25°C)

| TEST | Device types 01 and 02 | |
|---|---|---|
| | Limit | Delta |
| $V_{BE}$ | .600 V min<br>.800 V max | ±.010 V |
| $\lvert V_{BEQA} - V_{BEQB} \rvert$ | 2.0 mV | |
| $h_{FE}$ | 70 min<br>300 max 1/ | ±10% |

1/ The 300 max limit applies to device 01 only."

4.6, delete.

6.4, line 2: Delete "MIL-STD-1313, MIL-STD-1331, MIL-S-19500E," and substitute "MIL-M-38510, MIL-STD-1331, MIL-S-19500,".

PAGE 31

6.6, delete and substitute:

"6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-Industry type. Generic-Industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-Industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

| Military device type | Generic-Industry type |
|---|---|
| 01 | 3018A |
| 02 | 3045 " |

Note: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment

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